

Operation Uptime: questions Joveo's CEO is likely to ask, and the answers

Every number below comes from the delivered model (Operation_Uptime_Model.xlsx) or from a what-if run on a copy of it. "Show" names the cell or slide to put on screen. Sources are linked on the Assumptions tab, next to each input.

A. The headline

1. What exactly am I approving? €117.5k for 100 signed maintenance contracts by Fri 18 Dec. €99.8k is planned spend: €70.7k paid channels including €9k of tests, €27.4k for the three new sources, and €1.6k set-up. The other €17.6k is a reserve, released only by the Tracker rules from week 3.

- Excluded: Joveo's fees and the client's internal recruiter time.
- *Show:* Summary C5–C7, Budget C4–C12, slide 5.

2. How sure are you of 100? 100 if the funnel rates hold, with assessment slots 98% used (308 qualified for 314 slots).

- The risk is capacity, not money.
- Pass rates 10% worse → ~92 hires. 30% worse → ~71.
- The client can't do 55 a week and manages only 45 → ~83 hires.
- *Show:* Funnel B22–B26, slide 3 chart, slide 1 "How sure".

3. Is €998 per hire cheap or expensive? Cheap.

- **German benchmark:** total cost of a skilled hire is €4,700, of which about a third (~€1,570) is recruiting (IZA DP 7656, 1,001 firms, <https://docs.iza.org/dp7656.pdf>, 2013 euros).
- **Agency:** 20–30% of a €37–45k salary (gehalt.de) = €7,400–13,400 per hire, so €0.74–1.34M for 100.
- **US comparison:** about \$4,700 per hire (SHRM).

4. Why not spend more to be safe? More money buys at most ~2 more hires: there are only 6 spare assessment slots, and qualified needed ÷ hires is about 3.

- After that the 55-a-week limit binds.
- The real safety lever is +9 assessment slots a week, which brings use down to 85%, the occupancy level where Bagust et al. (BMJ 1999, <https://pubmed.ncbi.nlm.nih.gov/10406748/>) find risks start.
- *Show:* Budget B50, slide 1 asks.

5. Why a 15% reserve? Three reasons:

- High-risk projects carry 10–20% contingency (Mastt).
- The reserve plus rounding (€17.6k) covers about 40% of the extra media a 30% miss on the qualified share would need (+43%, about €43k).
- IZA finds the cost of each extra hire rises when a firm hires 20 or more at once.

It is released only from week 3 by the Tracker rules. With no reserve the budget would be €100k.

6. What do I get with 30% less (€82.25k)? About 87 hires.

- Electricians fall to ~21 of 30, supervisors to ~7 of 10.
- Mechatronics and automation stay almost full.

- Rule: cut the most expensive cost-per-hire cells first.
 - Advice: keep the money; if forced, cover supervisors by internal promotion and protect electricians.
 - Show: Budget rows 45–48, slide 6.
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B. Funnel and timeline

7. Why 308 qualified applicants for 100 hires? Qualified → signed = pass × offer × acceptance.

- Electricians: $0.50 \times 0.90 \times 0.70 = 31.5\%$.
- Blended across roles: 32%, so $100 \div 0.32 \approx 311$ needed. The plan delivers 308 qualified, which means 100 hires because the cheaper new sources convert better.
- Sources: pass rates anchored on Gem 2025; acceptance on Königsteiner/StepStone (71–78%); 25% of applications qualified from IAB 2026.
- Show: Funnel rows 5–9, slide 2.

8. Your 70–75% acceptance is far below SmartRecruiters' 91%. Too pessimistic? The 91–92% figure is a global benchmark. German surveys give 71–78%, so the plan uses those.

- If 91% held: budget €85k, slots only 77% used.
- That's upside, checked in week 4 once ≥ 7 offers are out (rule R10).

9. Why is the pass rate the most sensitive input? Because slots are full.

- A lower pass rate means more qualified applicants per hire, and there are no spare slots to assess them, so money can't buy the hires back: 30% worse → ~71 hires.
- A lower qualified share or a higher cost per application only costs money (+43% / +30% media), with hires unchanged.
- Show: Funnel rows 41–47, slide 3.

10. Why must applications arrive by 24 Nov? Work backwards from Fri 18 Dec:

- 7 days offer → signature
- 3 days interview → offer
- 8 days assessment → interview
- 6 days application → assessment

That gives 24 Nov. Supervisors add one extra round of 7 days, so their cut-off is 17 Nov.

18 Dec is the last date because job-search activity drops about 45% over the holidays (Indeed Hiring Lab DE).

11. German time-to-hire is 55 days. You assume 24? Yes. It's a sprint, and it depends on client commitments: offers within 3 days, e-signature, and offers valid for 7 days. These are in the asks.

- Each extra day of offer approval costs ~8 assessment slots.
- If offers take 7 days instead of 3: ~92 hires (model run).
- The 6 + 8 days for screening and interview scheduling are SmartRecruiters' Germany medians.

12. Is assessment the only bottleneck? What about interviewers? The brief gives assessment capacity only. In the model every qualified applicant who is assessed also takes the technical test or interview, so hiring managers need about 55 interview slots a week as well.

- Raise this as part of the "55 assessments/wk" ask.

C. Channels and Joveo

13. Programmatic is Joveo's core product. Why does it get only €8.7k and 4 hires? At the market-median Indeed cost of €161 per application (TalentBait, 2,549 DACH campaigns) and job-board quality, programmatic costs €2,317 per hire. That's more than StepStone (€1,564), LinkedIn (€1,687) or Meta (€1,963), so the model buys it last. Two points change the picture:

- **Joveo's platform runs LinkedIn, Meta and search too** ("Reach passive candidates across LinkedIn, Meta, search...", joveo.com), so most of the €70.7k of paid media can run through Joveo.
- **Joveo's own published manufacturing cases show cost per application down 17–32%.** Applied to programmatic, the budget falls to €112.5k–€107.5k and programmatic takes more hires (model runs).

These are vendor results, so they're shown as upside, not built into the base.

- *Show:* slide 4 footnote, Channels rows C05–C08.

14. Why LinkedIn for blue-collar roles? Only for automation and supervisors (PLC/TIA and Meister profiles).

- LinkedIn reaches 15% of skilled-worker job seekers (Trendence 2023, 42,817 job seekers).
- Its cap = reachable in-market pool × 15% ÷ qualified share.

15. Are Meta applicants any good? Lower quality: the social index is 0.50 of the average (IAB employer survey). The model already prices that in, at 12.5% qualified.

- Volume is capped at Redline's median of 20 qualified per Elektroniker campaign (181 trade campaigns).
- It still buys 6–7 electricians at ~€2,100 each.

16. Why spend €9k testing every cell? That's 9% of the budget, matching the common "10% testing budget" practice.

- Every test reaches its €750 switch-off point by 9 Nov, so each channel is judged on real data.
- Test applicants are real candidates and their hires count. Duds are cut from Mon 2 Nov.

17. How do you decide to move money each week? The Tracker rules, each sourced or statistical:

- **Scale up** if cost per qualified $\leq 0.75 \times$ plan (95% one-sided test).
- **Cut 50%** at $\geq 1.5 \times$, **switch off** at $\geq 2 \times$ (ad-industry rules).
- **Switch off** after €750 with zero qualified when ≥ 3 were expected (5% false-kill risk, Poisson).
- **Pause** the costliest cells if next week is forecast above 55 qualified; **release reserve** below 50.

The rules were tested: all 17 change a decision when actuals are typed in.

- *Show:* Tracker X column, slide 8.

18. 20 free organic hires. Realistic? Organic volume = openings × 27 applications per vacancy at firms with 250+ staff (IAB) × 13% coming via the careers page (CareerPlug). Qualified share = 25% × careers-page index 0.71 = 17.8%.

- It is free media, but it uses slots like everything else.

D. The new sources

19. Is re-contacting past applicants legal? Only consented talent-pool records, typically kept for up to a year.

- Day-1 ask: a GDPR-checked export.
- The pool is 211 records: $10,703 \text{ staff} \times 7.8\% \text{ in maintenance (BA occupation data)} \times 12\% \text{ turnover (IAB)} \approx 100 \text{ hires a year, } \times \text{runners-up per hire.}$
- At ~€14 a hire it's the cheapest source.
- If the export is empty, the budget rises to €130k.
- Joveo's AI Talent CRM does this re-engagement by email, SMS and WhatsApp, which makes it the repeatable product.

20. Is a €1,000 referral bonus enough? Most German firms pay €501–1,000 (Personalwirtschaft/Radancy, 335 firms); hard-to-fill roles get €2,000 (KOFA example).

- Referrals are sized to deliver 11% of hires (CareerPlug); the model gets 11 hires.
- The bonus is paid only on a signed hire, so it comes out of the budget.

21. Will construction electricians cut it? Only if the client accepts the profile. That decision is needed by 23 Oct.

- They face the automation-level pass rate (factor 0.9) and get a 2–4 week in-house bridge.
- 120 applications → 8.4 hires at €1,329.
- If the client says no, programmatic ads fill all 30 electricians and the budget rises to €127.5k.

22. Polish corridor: legal, tax, language?

- **Legal:** EU free movement, so no work permit.
- **Insurance and tax:** social insurance in the country of work (TK); German working days taxed in Germany, with no special commuter rule in the Germany–Poland treaty.
- **Recognition:** mechatronics is non-regulated, so none is needed.
- **Language** is the binding filter: 14% of vocationally trained Poles speak a foreign language well (Eurostat AES 2022). A Polish-speaking freelancer pre-screens applicants.

It delivers 18 hires (the 40% cap) at €383 each. Without it the budget would be €150k.

23. Why not foreign-trained electricians? The brief says recognition takes 2–4 months, so the plan buys none. But in German law the industrial electrician is non-regulated for employees (BIBB: “This profession is not regulated in Germany”), so recognition is legally needed only for self-employment or the master-craftsman route.

- If the client accepts employer-verified competence for its Elektrofachkraft appointment, EU-trained electricians become an extra route before 31 Dec.
- That's an upside to raise.

24. Why not just use an agency or temp workers?

- **Agency:** €7,400–13,400 per hire, 6–11× this plan.
- **Temp:** billed at about 2× wage, capped at 18 months per worker, equal pay after 9 months. Taking a temp on permanently costs a fee of up to ~2 monthly salaries.

Temps work as a bridge for a few critical shifts, not as the plan.

E. Risk and operations

25. What is the single biggest risk? Assessment capacity: 98% used.

- **Mitigations:** pre-screened sources first (rediscovery, Polish pre-screen calls), weekly pacing ≤ 55 , weekly “Full/OVER” flags, and an ask for +9 slots a week.
- With 64 slots a week, use falls to 84%.

26. What if week-1 pass rates come in low? Once ≥ 10 interviews are in (NIST $n \cdot p \geq 5$ rule), if the pass rate is below plan (electricians 50%):

- tighten knock-out questions
- add pre-screen calls
- ask for more slots
- re-run the model

10% worse $\approx$ 92 hires.

27. Why is electrician the hardest role?

- **Scarcest trade in Brandenburg:** 85.7% of vacancies have no suitable unemployed candidate, vs 41.4% across all jobs (KOFA 2026).
- **Most money:** 38% of spend.
- **No alternative route** under the brief: no Polish route and no foreign-trained route.

Supervisors cost more per hire (€1,550) but are only 10 people. Automation has thinner channel coverage but only 15 hires.

28. Doesn't promoting a technician to supervisor just create another opening? Yes. That's why internal promotion is only the fallback for a supervisor shortfall, not a source in the plan. Each promotion adds a technician backfill.

29. How will you track and attribute?

- **Data:** tracked URLs with utm tags captured in the applicant system's source field; a Monday export; recruiter disposition codes; the pre-screener sheet.
- **Tracker tab:** blends actuals with plan (30-application prior) and outputs scale/cut/off decisions plus a hire forecast.

30. What does cash flow look like?

- Weeks 1–2: about €34k (tests, StepStone ads, set-up).
- Weeks 3–6: €14–16k a week.
- Weeks 7–9: €2k a week of referral bonuses as people sign.
- Paid media stops 24 Nov.
- *Show:* Weekly M/N, slide 5 chart.

31. What will we know by 9 Nov?

- pass rate on the week-1 cohort
- each test cell's cost per application and qualified share
- size of the applicant-system export
- the client's construction-electrician decision
- first reads on the Polish corridor and construction-electrician tests (pass/kill bars on Initiatives rows 53–55)

F. Can I trust the model?

32. How do I know the spreadsheet is right?

- An independent Python re-implementation recomputes 59 outputs.
- 153 scenarios pass, including edge cases.
- Each of the 183 inputs was changed one at a time with no errors and matching results; derived cells recalculate live from them.
- All 17 Tracker rules change a decision when actuals are typed in.
- The deck's 45 headline figures match the model.
- Report: `tests/README.md`.

33. Which numbers are guesses? None are unexplained. Every input is one of three kinds:

- **A sourced fact** with a link: costs, rates, pools, reach, timings, volumes.
- **A live formula** on those facts, shown in black text: channel caps, rediscovery pool, referral volumes.
- **A stated rule or client commitment:** decision thresholds based on NIST/BMJ/ad-industry conventions, the 3-day offer, the 18 Dec cut-off.

34. Aren't vendor numbers biased?

- TalentBait figures are medians across 2,549 campaigns.
- Joveo's case results are used only as upside, not in the base.
- Where vendor and neutral sources disagree, the model takes the neutral or more conservative one. For example, it uses the German 71–78% acceptance rather than SmartRecruiters' global 91%.

35. If I change a number live, will the model follow? Yes. Every derived input is a live formula. For example:

- Change the qualified share, active-searcher share or reach → caps and costs move.
- Change openings → referral and rediscovery volumes move.
- Change the lags → the cut-off and StepStone ad counts move.

G. Joveo's business

36. Where are our fees? Not in the €117.5k. Joveo prices custom (media-based, volume discounts, a performance guarantee), so the fee is quoted separately.

37. Is this repeatable for other clients? Yes, through three pieces:

- **Applicant-archive rediscovery** using Joveo's AI Talent CRM.
- **Tracked-to-hire attribution** using utm source tags.
- **A planning model driven by public statistics** (BA, IAB, Eurostat), so it adapts to any German region and trade by changing the inputs.